

Environmental Technology

ISSN: 0959-3330 (Print) 1479-487X (Online) Journal homepage: www.tandfonline.com/journals/tent20

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To cite this article: Juan Wang, Xindong Wang, Chunlin Bian, Jianguo Liu & Benyi Xiao (2025) Effect of aeration pretreatment on anaerobic digestion of swine manure, Environmental Technology, 46:12, 2035-2051, DOI: 10.1080/09593330.2024.2416094

To link to this article: https://doi.org/10.1080/09593330.2024.2416094

Published online: 16 Oct 2024.

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ENVIRONMENTAL TECHNOLOGY
2025, VOL. 46, NO. 12, 2035–2051
https://doi.org/10.1080/09593330.2024.2416094

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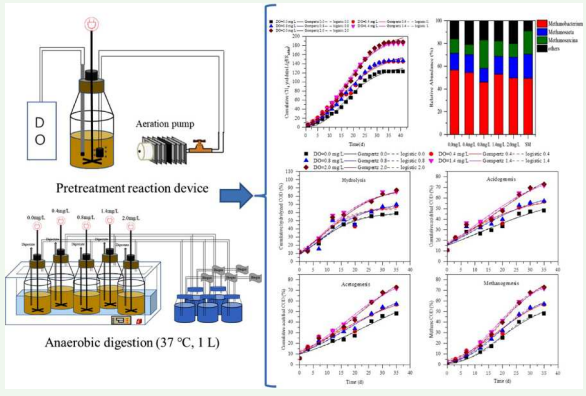
Effect of aeration pretreatment on anaerobic digestion of swine manure

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ABSTRACT

To investigate the effects of aeration pretreatment on the anaerobic digestion (AD) of swine manure, five pretreatment groups were established with dissolved oxygen (DO) in each group set to 0.0, 0.4, 0.8, 1.4, and 2.0 mg/L, respectively. The results demonstrated that compared to the non-aeration group, methane production increased to varying degrees with different aeration pretreatments (AP), with a maximum increase of 27.98% (DO = 2.0 mg/L). AP reduced the hydrogen sulfide (H_2S) content of biogas. The H_2S concentration in the DO = 2.0 mg/L was only 0.209%, and this represented an increased H_2S removal rate of 49.27% compared to that of the DO = 0.0 mg/L (0.412%). Simultaneously, AP increases the hydrolysis rate. When the DO concentration reached 2.0 mg/L, the hydrolysis rate reached its maximum. An increase in the hydrolysis rate further enhanced the removal rate of organic matter. The organic matter removal rate was highest (36.96%) at DO = 2.0 mg/L. AP effectively prolonged the methane generation time and shortened the lag time of methane generation. AP creates a brief micro aerobic environment, accelerates substrate hydrolysis, and promotes the production and consumption of total volatile fatty acids, particularly acetic acid. Additionally, AP promoted the symbiotic relationship between *Caldicoprobacter* (20.93%–34.96%) and *Metanosaeta* (14.73%–18.45%).



ARTICLE HISTORY
Received 2 July 2024
Accepted 2 October 2024

KEYWORDS
Aeration pretreatment; anaerobic digestion; hydrolysis; kinetic analysis; swine manure

1. Introduction

With increasing demand for meat, eggs, and dairy products, the livestock industry has developed rapidly in recent years. Pork is a major source of human food, accounting for over 40% of the total meat consumed worldwide [1]. The rapid development of pig farming has produced large amount of manure, and swine manure (SM) contains large amounts of organic matter, nitrogen, and phosphorus. If not handled and stacked properly, it can cause environmental pollution, affect human health, and can easily cause eutrophication of water bodies when discharged into water [2,3]. Anaerobic digestion (AD) provides a new method for treating SM and is an important technology for organic waste treatment and biomass energy utilisation. AD protects the environment and reduces pollutants while reducing the use of fossil energy using the produced biogas as a fuel, and the digestate can be used as an organic fertiliser for agricultural production, thus facilitating the recycling of resources and energy [4].

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Despite the merits mentioned above, AD of SM possesses several drawbacks, such as a low hydrolysis rate, toxic hydrogen sulfide (H_2S) release, and poor system stability. Previous studies demonstrated that microaeration can overcome these issues. Traditionally, AD systems have been maintained in a strictly anaerobic environment, as oxygen inhibits the activity of specific anaerobic microorganisms and methanogens [5]. However, it was recently reported that introducing a small amount of air or oxygen into the digester can accelerate the AD reaction rate, increase methane yield, promote substrate degradation, and reduce the H_2S content of biogas [6,7]. However, due to its sensitivity to changes in equipment and internal material conditions [8,9], direct long-term and excessive oxygenation also damages the stability of anaerobic systems, ultimately leading to higher energy consumption. Aeration pretreatment (AP) of AD raw materials can effectively avoid these problems. As AP is performed outside the digester, it is often more convenient, does not affect the activity of methanogens, and does not damage the airtightness of the digester.

Previous studies have investigated the effect of AP on the AD of sewage sludge and olive factory wastewater and observed that AP can enhance the hydrolysis of organic matter, increase the methane yield and organic matter removal in AD, boost enzyme activity, and reduce the concentration of H_2S. Rashvanlou et al. [10] observed through AD of sewage sludge that AP (0.2 L_{air}/L_{liquid}/min [vvm] for 40 h) increased the soluble chemical oxygen demand (SCOD) by 65% and key enzyme activity by 10%. Jenícek et al. [11] conducted AD of municipal sludge in an engineering application (3,000 m^3) and observed that AP (1.2 m^3/h) could remove up to 99.5% of the H_2S from biogas. Alrawashdeh et al. [12] reported that AP (0.5 vvm for 4 days) can improve the removal rates of volatile solids (VS) (56%) and total chemical oxygen demand (TCOD) (70%) and increase biogas yield by 111% during the AD of olive factory wastewater. Meanwhile, to optimise AP, previous studies have also investigated the effect of AP parameters such as intensity, time, and temperature on AD [13]. For example, Montalvo et al. [13] compared the effects of different AP Aintensities, time, and temperatures on AD of sewage sludge and observed that AP increased the concentrations of soluble proteins, soluble polysaccharides, and SCOD, and improved the removal rate of VS with the optimal AP parameters as 0.3 vvm, 48 h, and 35 °C. Although most previous studies used the aeration rate (vvm) to represent AP intensity, the dissolved oxygen (DO) level was not determined, and this approach cannot reveal the DO level of the pretreated solution and the actual oxygen concentration that contacts the organic matter and microorganisms. Aeration rates often are different for different equipment and environments. Additionally, it remains unclear whether AP affects biochemical processes other than hydrolysis in AD, and the study of dynamic models remains in the exploratory stage [14].

Therefore, this study explored the effect of AP on the AD of SM by setting different DO levels. Dynamic models were used to reveal the effects of AP on the four main AD steps (hydrolysis, acidogenesis, acetogenesis and methanogenesis). It hopes that this study provides a reference for the improvement of AD in the context of SM.

2. Material and methods

2.1. Feedstock and inoculum

The SM used in this study was obtained from a pig farm in Beijing, China, and the inoculum was sludge obtained from a laboratory anaerobic digester. All materials were sampled on site, and the manure was hand-picked to remove visible weeds and gravel particles. It was mixed well, and some samples were then used to test their basic physicochemical properties. The remaining samples were stored in a refrigerator at 4 °C. AD sludge was used as the inoculum, and its basic physicochemical properties were tested. The basic physicochemical properties of the experimental materials are presented in Table 1.

2.2. Aeration pretreatment

The AP of SM was carried out in a 1 L reactor (working volume 0.65 L). SM and inoculum were first mixed at a ratio of 1:3 (v/v) in the reactor, and the pH of the mixture was adjusted to 7.5 ± 0.1 using 0.1 M HCl. A 1.5 W aeration pump (AquaBlue, China) with 0.3 vvm was then used, and a DO meter was used to monitor

Table 1. Physicochemical characteristics of swine manure and inoculum.

Items	Swine manure	Inoculum
TS	31.85 ± 0.27	4.23 ± 0.06
VS	24.53 ± 0.21	2.57 ± 0.07
pH	7.52 ± 0.09	7.88 ± 0.09
Carbon content (%TS)	39.23 ± 0.61	ND
Nitrogen content (%TS)	3.64 ± 0.21	ND
C/N	11.08 ± 0.33	ND
TCOD (g/L)	159.10 ± 0.65	ND
T-Carbohydrate (g/L)	23.21 ± 0.32	ND
T-Protein (g/L)	49.45 ± 0.47	ND

Note: Data include the means ± standard deviation ($n = 3$), ND: Not detected, TS: total solids, VS: volatile solids, TCOD: Total chemical oxygen demand; T-Carbohydrate: Total carbohydrate; T-Protein: Total protein.

Table 2. Characteristics of substances inside the reactor after AP.

DO level (mg/L)	0.0	0.4	0.8	1.4	2.0
pH	7.60 ± 0.05	7.64 ± 0.03	7.65 ± 0.06	7.67 ± 0.07	7.68 ± 0.05
TS(g/L)	64.71 ± 0.11	63.83 ± 0.26	63.61 ± 0.09	62.62 ± 0.23	61.85 ± 0.18
VS(g/L)	52.77 ± 0.05	51.52 ± 0.19	50.48 ± 0.07	48.95 ± 0.14	47.21 ± 0.16
TCOD(g/L)	88.30 ± 0.50	87.50 ± 0.40	87.00 ± 0.50	86.10 ± 0.30	85.40 ± 0.60
SCOD(g/L)	25.90 ± 0.10	26.70 ± 0.30	27.80 ± 0.30	29.50 ± 0.20	31.10 ± 0.10
T-protein(g/L)	38.21 ± 0.29	36.79 ± 0.34	35.47 ± 0.18	33.46 ± 0.36	32.11 ± 0.24
T-carbohydrates (g/L)	13.19 ± 0.08	12.03 ± 0.12	10.89 ± 0.20	9.12 ± 0.14	8.49 ± 0.04
TAN(g/L)	0.83 ± 0.07	0.71 ± 0.08	0.64 ± 0.05	0.46 ± 0.03	0.38 ± 0.01
TVFA(g/L)	6.98 ± 0.16	6.12 ± 0.12	5.49 ± 0.15	4.45 ± 0.07	3.89 ± 0.03

SCOD: Soluble chemical oxygen demand; TAN: Total ammonia; TVFA: Total volatile fatty acids.

the DO concentration in the reactor. Five DO levels (0.0, 0.4, 0.8, 1.4, and 2.0 mg/L) were used. The aeration time of each AP was 0.5 ± 0.08 h (DO = 0.4 mg/L), 0.9 ± 0.14 h (DO = 0.8 mg/L), 1.7 ± 0.25 h (DO = 1.4 mg/L), and 2.3 ± 0.33 h (DO = 2.0 mg/L). For DO = 0.0 mg/L, the reactor was purged with pure nitrogen (99%) for 5 min to remove the DO. After the DO level reached the set value, the reactor was intermittently aerated for half an hour to ensure that the DO level remained stable at the set value. After AP, the reactors were immediately sealed. Table 2 summarises the physicochemical properties of the substances in the reactor after AP.

2.3. Batch anaerobic digestion

Batch AD was conducted in laboratory-scale testing to evaluate the effect of APs (DO = 0.0, 0.4, 0.8, 1.4, and 2.0 mg/L) on the CH_4 yield of SM. The digesters were placed in a constant temperature water bath at 37±1 °C and stirred at 150 rpm. The batch ADs lasted for 36 days.

2.4. Chemical analysis

Biogas production was measured by collecting the drainage gas. The methane content of the biogas was detected using a gas chromatograph (GC) (Agilent 6890N, USA) equipped with a flame ionisation detector (FID) and a 30 m capillary column (HP-5). The H_2S content of the biogas was determined using a GC (Agilent 6890N, USA) equipped with an FID and a 30 m capillary column (DB-1701). The total sulfur content of the digestive products was monitored using an organic element analyzer (Elementar Rapid cs Cube, Germany). Sulfur-containing samples were placed in quartz combustion tubes using an automatic sampler. Under the catalysis of tungsten trioxide oxidation at a high temperature of 1150 °C and the driving force of oxygen carrier gas, all sulfur elements were converted into sulfur dioxide. After drying, it entered an infrared detector, and the sulfur content was determined by monitoring the sulfur dioxide concentration. Another GC (Agilent 6890N, USA) equipped with an FID was used to detect the volatile fatty acids (VFAs). The total and soluble indices of the digestate samples were determined. The samples for the analysis of soluble items were centrifuged at 8,000 rpm for 30 min and then passed through 0.45 μm filters prior to analysis. The digestate pH was measured using a pH meter (BANTE900, Shanghai Special Instruments Factory). Total solids (TS), volatile solids (VS), proteins, carbohydrates, ammonia nitrogen (TAN), and COD of the digestates samples were determined according to standard methods [15]. The digestates were sampled at 36 days to analyze the microbial community (bacteria and archaea) using a method similar to that used in previous studies [16].

2.5. Calculations and statistical analysis

Excel 2016 was used for data organisation and Origin 2019 was used for plotting. Dynamic analysis was performed using Origin 2019 software.

Experimental data regarding methane production were simulated in batch experiments using improved Gompertz [17] and logistic models [18] to obtain the kinetic parameters. The equations for these two models are as follows:

$$P(t) = P_0 \cdot exp\left\{-exp\left[R_{max} \cdot e \cdot \frac{(\lambda - t)}{P_0} + 1\right]\right\} \quad (1)$$

$$P(t) = \frac{P_0}{\left\{1 + exp\left[\frac{4R_{max}(\lambda - t)}{P_0} + 2\right]\right\}} \quad (2)$$

$$T_{ef} = T_{90} - \lambda \quad (3)$$

where P(t) is the cumulative methane yield (L/kg VS); P_0 is the potential methane yield (L/kg VS); R_{max} is the maximum methane yield (L/kg VS/d); λ is the lag phase (d); T is time (d); *e* was the expression (1) = 2.7183; T_{ef} is the effective methane yield time (d); T_{90} is the technical digestion time (d), indicating the time required to reach 90% of the maximum cumulative methane yield.

The reaction rates of the four AD steps (hydrolysis, acidogenesis, acetogenesis, and methanogenesis) were expressed as COD conversion rates and were calculated as follows [19]:

$$R_{Hyd.} = \left(\frac{V_{CH_4}}{350} + C_{SCOD} \cdot V\right) \cdot COD_{addad}^{-1} \times 100\% \quad (4)$$

$$R_{Acid.} = \left(\frac{V_{CH_4}}{350} + C_{TVFA} \cdot V\right) \cdot COD_{addad}^{-1} \times 100\% \quad (5)$$

$$R_{Acet.} = \left(\frac{V_{CH_4}}{350} + C_{acetate} \cdot V\right) \cdot COD_{addad}^{-1} \times 100\% \quad (6)$$

$$R_{Meth.} = \frac{V_{CH_4}}{350} \cdot COD_{addad}^{-1} \times 100\% \quad (7)$$

where $R_{Hyd.}$, $R_{Acid.}$, $R_{Acet.}$, and $R_{Meth.}$ are cumulative hydrolysed COD (%), cumulative acidified COD (%), cumulative acetified COD (%), and methane COD (%), respectively; V_{CH4} is the volume of methane (mL); V is the volume of working (L); C_{SCOD} is the concentration of soluble COD (g/L); C_{TVFA} is the concentration of total volatile fatty acids (TVFA) (g COD/L); $C_{acetate}$ is the concentration of acetate acid (g COD/L); COD_{added} is the COD of substrates added into the bottles (g), and 1 g of COD is equal to 350 mL CH_4.

Equations (2) and (3) were also used for reaction rates of the four AD steps. In this case, the parameters are represented as follows: $P(t)'$ is the conversion ratio of substrates (%); P_0' is the conversion potential (%); R_{max}' is the maximum conversion rate (%/d).

This study employed nonlinear regression analysis and Origin Pro 2016 software to fit the curves and estimate the dynamic parameters of the modified Gompertz and logistic models [20]. The adaptability of the model was verified using experimental data obtained from all digesters. Statistical indicators and correlation coefficients (R^2) were utilised [21].

The energy consumption of AP was calculated as follows [14]:

$$E_{AP} = \frac{3600 \cdot P \cdot t_{pretreatment}}{V \cdot C_{TS}} \quad (8)$$

where, E_{AP} is the energy consumption of the AP based on the added unit SM (J/g TS), P is the power consumed by the AP (W), $t_{pretreatment}$ is the AP time (h), V is the volume of the reactor (L), and C_{TS} is the total solids of the SM (g/L).

Energy Output of AD was calculated as follows [14]:

$$E_0 = Y_{CH_4} \cdot \varepsilon \cdot \lambda_A \quad (9)$$

where Y_{CH4} is the methane yield based on the added unit SM (L CH_4/g TS), ε is the lower heating value of methane (35,800 J/L CH_4), and λ_m is the energy conversion factor of methane (0.9).

Redundancy analysis (RDA) was used to evaluate the relationships between microbial communities, environmental factors, and production. RDA was conducted using the CANOCO 4.5 package [22]. Statistical significance was tested using analysis of variance, and the means were separated using Tukey's honest significant difference test with a significance threshold of 0.05 [23].

3. Results and discussion

3.1. Gas indicators

3.1.1. Biogas production

In the AP AD experiments, various gas yield indicators of each group were compared to those of the control group to provide a comprehensive analysis of the effect of different oxygen contents on digestion. The daily gas yield, cumulative gas yield, methane content, and cumulative methane yield of the digestion units with different oxygen contents are presented in Figure 1.

In the AD experiment with AP, the DO level exerted a significant impact on gas yield and reaction stability throughout the AD process. Overall, the main purpose of the pretreatment was to increase the methane yield, accelerate the AD process, and improve the substrate degradation rate. Cumulative methane yield was calculated from the first day of AD until no significant gas yield was detected. From Figure 1(a) it could be observed that the gas yield of the five groups of the experiment rose faster on the first day, among which, the gas yield of the DO = 0.4 mg/L group reached 9.53 mL/g VS/d. This was followed by the DO = 0.8 mg/L group with a gas yield of 7.72 mL/g VS/d, likely due to the heat expansion of the gas in the digestion tank and the increase in gas pressure in the digestion tank that primarily represented the hydrolysis stage of the AD process at this time. The concentration of methane in the produced gas was low, as presented in Figure 1 (c). The methane content increased rapidly after day 5, when the more easily hydrolysed sugars in the digester were partially decomposed and utilised by anaerobic bacteria and when methanogens began to grow. The reaction began, and the daily gas yield of each group gradually increased, reaching its peak on the 23rd day before rapidly decreasing thereafter. Daily methane yield exhibited the same trend. The small amount of gas produced in the early stages of the reaction may be related to the activity of the inoculum, which also possesses the potential to increased gas yield. As the reaction progressed, a large amount of VFAs was produced during the acidification stage, and the gas yield

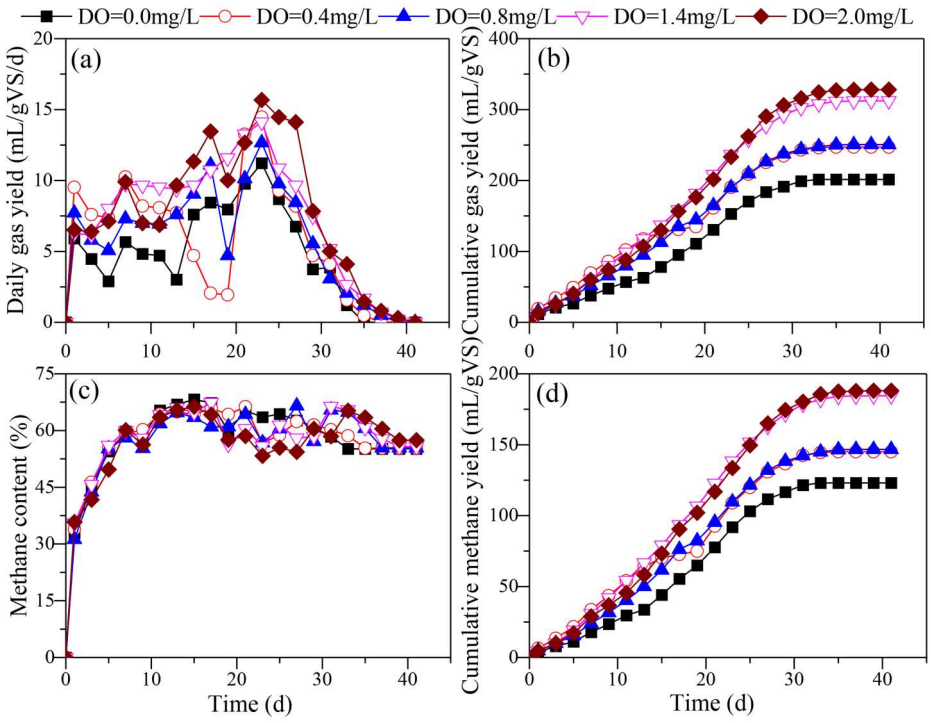


Figure 1. (a) Daily gas yield with time; (b) Cumulative gas yield with time; (c) Methane content; (d) Cumulative methane yield with time.

increased further. According to the gas yield, only the DO = 0.4 mg/L group exerted an inhibitory effect during the reaction process. After the device stopped running, the cumulative methane yield of five different AP groups were 123.10 (DO = 0.0 mg/L), 145.07 (DO = 0.4 mg/L), 146.48 (DO = 0.8 mg/L), 184.05 (DO = 1.4 mg/L), and 187.47 mL/g VS (DO = 2.0 mg/L), respectively. Four different DO levels increased the methane yield of SM by 17.85% (DO = 0.4 mg/L), 18.99% (DO = 0.8 mg/L), 25.65% (DO = 1.4 mg/L), and 27.98% (DO = 2.0 mg/L) compared to that of the control group (Figure 1(d)). These results indicate that after AP of the SM, the methane yield could be significantly improved. By comparing the effects of different DO levels on methanogenesis, it was obsered that gas yield gradually increased with increasing DO concentration. However, after the DO concentration reached a certain level, its effect on the gas yield gradually decreased. For example, the DO = 2.0 mg/L group exhibited the highest cumulative methane yield but only increased by 1.86% compared to that of the DO = 1.4 mg/L group. The results indicated that the impact of different DO levels on the methane yield in the AD of SM varied. This may be due to the observation that all APs enhance the hydrolysis of organic matter, boost enzyme activity, and reduce the inhibition of H_2S, TAN, and VFAs by lowering their concentrations with different effects [24–27]. Therefore, when aeration is used as a pretreatment method for the AD process, the amount of aeration must be strictly controlled, as excessive oxygen may inhibit the activity of methanogens and further reduce the methane yield, even affecting the stable operation of the AD system [26]. Moreover, excessive oxygen might oxidise certain substances required by methanogens (such as VFAs) or aerobic methane oxidising bacteria might consume methane.

The energy consumption of each AP and the energy production for each reactor are summarised in Table 3. The results revealed that the energy consumption of AP increased from 65.08 J/g TS to 308.84 J/g TS when the DO levels increased from 0.4 mg/L to 2.0 mg/L. Meanwhile, the energy production increased from 2,541.06 J/g TS to 4048.65 J/g TS when the DO levels

Table 3. Energy consumption of AP and energy production for each reactors.

DO level (mg/L)	Energy consumption of AP (J/g TS)	Energy production (J/g TS)	Energy Output (J/g TS)	Increase (%)
0.0	–	2541.06	2541.06	–
0.4	65.08	3036.01	2970.93	16.92
0.8	117.54	3075.96	2958.42	16.42
1.4	225.54	3925.98	3700.44	45.63
2.0	308.84	4048.65	3739.81	47.18

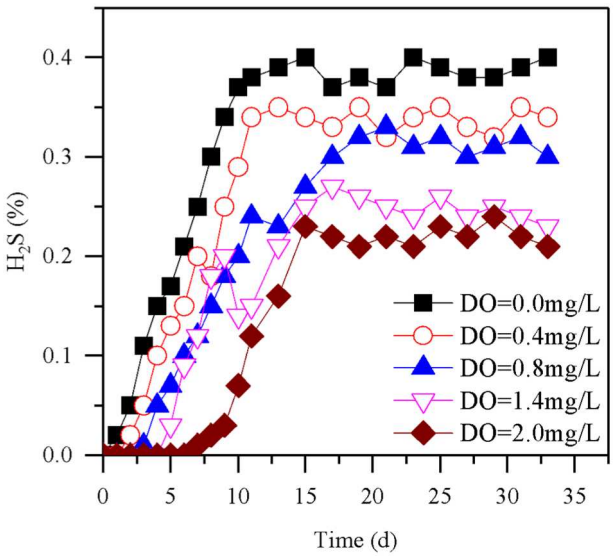


Figure 2. H_2S content in the test.

increased from 0.0 mg/L to 2.0 mg/L. Therefore, the energy output of AD increased from 2,541.06 J/ g TS to 3,739.81 J/ g TS when the DO levels of the AP increased from 0.0 mg/L to 2.0 mg/L. Compared with the DO = 0.0 mg/L group, the APs increased the energy output by 16.92% (DO = 0.4 mg/L), 16.42% (DO = 0.8 mg/L), 45.63% (DO = 1.4 mg/L), and 47.18% (DO = 2.0 mg/L), with the DO = 2.0 mg/L group being the highest. These results indicated that AP could increase the energy output of the AD of the SM.

3.1.2. H_2S content

H_2S formation is a challenging problem that occurs during AD, and the effects of different DO levels on H_2S formation were assessed (Figure 2). In the DO = 0.0 mg/L group, the H_2S concentration increased sharply at the beginning of the reaction. As the DO level increased, the appearance of H_2S in the biogas exhibited varying degrees of delay. The H_2S production time in the DO = 0.4, 0.8, 1.4, and 2.0 mg/L groups was delayed by 1, 2, 4, and 6 days, respectively, compared to that of the control group. At the end of the reaction, the H_2S content in each group of biogas remained stable at 0.412% (DO = 0.0 mg/L), 0.349% (DO = 0.4 mg/L), 0.306% (DO = 0.8 mg/L), 0.234% (DO = 1.4 mg/L), and 0.209% (DO = 2.0 mg/L), respectively. Compared to the DO = 0.0 mg/L group, AP reduced the H_2S content from 15.29% (DO = 0.4 mg/L) to 49.27% (DO = 2.0 mg/L), with the DO = 2.0 mg/L group exhibiting the best H_2S removal effect. The results revealed that the concentration of H_2S in the biogas decreased as the DO levels increased, indicating that oxygen effectively reduced the H_2S content in the biogas. This was similar to the results of previous research [28]. Sulfate reducing bacteria (SRB) reduce methane yield by competing for various organic and inorganic substrates (acetate, butyrate, fatty acids, hydrogen, and propionate) [29]. At pH 7–8, the generated H_2S is toxic to methane-producing bacteria, thus further inhibiting methane yield [24]. Therefore, the removal of H_2S reduces the inhibition of H_2S on methanogens and increases methane yield [30].

The effects of different DO levels on the concentration of sulfides in the matrix were different. The total sulfur concentration in the digested material was measured at the beginning and end of the test (Table 3). The results indicated that when DO = 0.0 mg/L, the total sulfur concentration in the termination sample was the lowest. As the DO levels increased, the total sulfur concentration in the digester also increased, ultimately reaching a maximum value of 4.445 mg/g in the DO = 2.0 mg/L group. AP exerts a significant impact on the biological and chemical processes of sulfides in substrates. During the traditional AD process, SRB reduced sulfur compounds in the substrate to H_2S in the form of biogas and sulfides in the liquid phase (HS^-, S^{2-}). H_2S is the product of sulfur reduction by the SRB. The addition of trace amounts of oxygen to the digester directly inhibited sulfate reduction and reduced the production of H_2S. In contrast, SRB were inhibited after AP, whereas sulfate-oxidising bacteria (SOB) proliferated in large numbers, further retaining sulfur-containing compounds in the biogas slurry. The overall biological desulfurisation reaction in the presence of SOB is as follows [30]:

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \quad (10)$$

$\Delta G^0 = -169.35$ kJ/mol

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+ \quad (11)$$

$\Delta G^0 = -732.35$ kJ/mol

$$2HS^- + 2O_2 \rightarrow S_2O_3^{2-} + H_2O \quad (12)$$

$\Delta G^0 = -387.35$ kJ/mol

Assuming that the H_2S concentration in the DO = 0.0 mg/L group was the maximum value produced by AD of SM, the oxygen required to remove all H_2S could be calculated based on chemical Equations (10)–(12). According to the calculations, the reduction of 3 mol of H_2S was expected to consume 7 mol of oxygen. Therefore, to remove all H_2S in the DO = 0.0 mg/L group, 0.116 g of oxygen was required. To remove all H_2S from the SM with TS = 63.71 g/L, a DO concentration of 116 mg/L was required.

Based on the H_2S yield, it could be concluded that the DO = 2.0 mg/L group was not the optimal oxygen

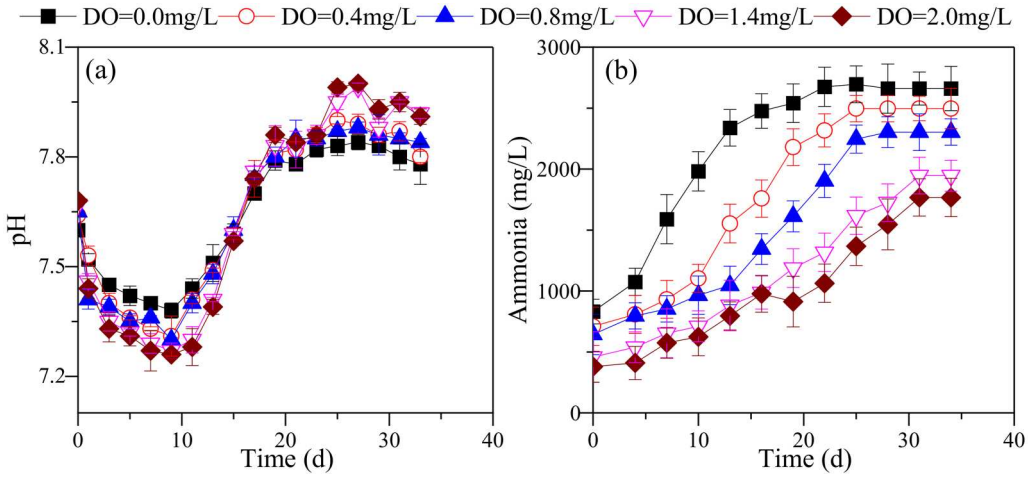


Figure 3. Changes in pH and TAN over time under different AP conditions.

concentration for the AD of SM through AP. However, based on the results of this experiment and considering the methane yield and economic benefits of each group, 1.4 mg/L was the optimal DO level for the AD of SM.

3.2. Liquid indicators

3.2.1. Effect on pH and TAN

Samples were taken every two days during the reaction to measure the pH (Figure 3(a)). The low concentration of inoculum and low numbers of methanogens during the first 4 days of the reaction process resulted in limited use of the VFAs produced by acid-producing bacteria hydrolysis and led to an accumulation of VFAs, at which time the pH of the reaction system decreased, with the lowest pH in the DO = 2.0 mg/L group falling to 7.26 on day 9. However, as the high nitrogen content of SM was supplemented with NH_4^+-N formed by hydrolysis, which could neutralise the acid produced to some extent, the pH drop at the beginning of the reaction did not inhibit the growth of methanogens. Thereafter, as the methanogens grew and multiplied, their ability to use VFAs to produce methane increased, and the daily methane concentration and gas yield increased. At this time, the pH rose rapidly, after which the VFAs produced by hydrolysis and acidification no longer accumulated until the end of the reaction, when the pH was finally stabilised at approximately 7.90. The pH remained stable between 7.26 and 7.92 throughout the reaction.

During AD, nitrogen is an important nutrient for microbial growth and reproduction, and the TAN exhibits a buffering capacity in the fermentation system. However, excessive nitrogen content in anaerobic fermentation substrates can inhibit microbial activity and reduce the biogas yield. In the AD of SM, the inhibitory concentration of TAN was 3,500–3,800 mg/L, and the process was blocked at 2,300–2,400 mg/L [25]. Figure 3 (b) show the changes in TAN inside each digester under different AP conditions. Different DO levels exerted a significant impact on the changes in TAN during the AD of SM. After the AP, the TAN concentrations of mixtures were reduced from 831.29 mg/L to 379.40 mg/L when the DO levels increased from 0.0 mg/L to 2.0 mg/L. During AD, the TAN concentration gradually increased and stabilised at the end of the test (Figure 3). The results demonstrated that the DO level of APs affected the TAN concentration with a higher DO level and a lower concentration of TAN, indicating that the DO levels reduced the inhibition of TAN (particularly free ammonia; FAN) and enhanced methane yield.

AP positively impacts the biological and chemical processes that alleviate ammonia inhibition in the AD of SM. In an aqueous solution, TAN exists in two forms including NH_4^+-N and FAN. AP significantly enhanced the early degradation of proteins and released NH_4^+-N and FAN into the SM. NH_4^+-N, as a buffering substance, could effectively prevent acid inhibition during AD. Ni et al. [31] demonstrated that appropriate AP in AD could improve organic matter removal efficiency and provide a buffer to neutralise the excess VFAs produced. FAN is the main ammonia component inhibiting anaerobic fermentation. As NH_3 is prone to volatilisation in aqueous solutions, gas (carrier gas) was introduced into the fermentation solution to fully contact the fermentation broth and allow NH_3 to pass through the gas–liquid interface and transfer to the gas phase, thereby achieving the removal of TAN.

3.2.2. Effect on COD

The trends in TCOD and SCOD content changes in the five digesters are presented in Figure 4. The TCOD

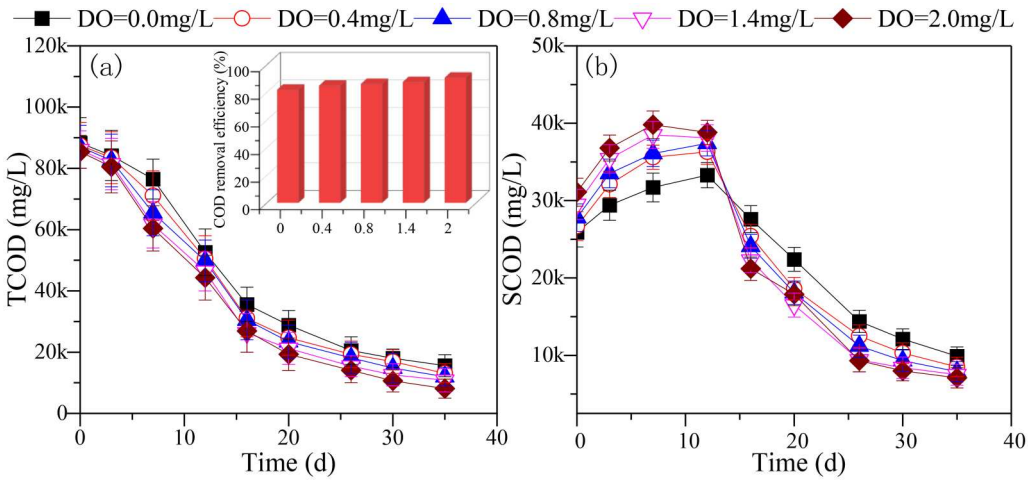


Figure 4. Changes in TCOD and SCOD over time under different AP conditions.

content in each digester continued to decrease as presented in Figure 4(a). As the DO levels in the digesters increased, the degradation rate of TCOD gradually increased. At the end of the reaction, the TCOD contents in each digester were 15,600 (DO = 0.0 mg/L), 13,100 (DO = 0.4 mg/L), 12,000 (DO = 0.8 mg/L), 10,800 (DO = 1.4 mg/L), and 8,100 mg/L (DO = 2.0 mg/L), respectively. Compared to the substrate, their COD removal rates ranged from 81.92% (DO = 0.0 mg/L) to 90.61% (DO = 2.0 mg/L) (Figure 4). Compared to the DO = 0.0 mg/L group, AP increased the COD removal rate from 3.54% (DO = 0.4 mg/L) to 10.61% (DO = 2.0 mg/L), with the DO = 2.0 mg/L group being the highest. The results indicate that AP at all DO levels increased the TCOD removal of AD, and different DO levels exerted different effects, with higher DO levels exerting a greater effect. Oliveira et al. [32] reported that AP (0.17 L_{O2}/L_{feed}/d) increased the removal efficiencies of VS and TCOD by 85.0% and 83.8%, respectively, during the AD of swine wastewater. These results indicate that AP allows for a higher TCOD removal rate in AD. AP exerted a significant impact on the biological and chemical processes involved in the removal of TCOD from the AD of SM. The explanation for this phenomenon is that AP promotes the hydrolysis of organic matter, that is difficult to degrade in an anaerobic environment and improves the removal rate of organic matter.

The SCOD concentrations in the DO = 1.4 mg/L group and the DO = 2.0 mg/L group reached their peaks on the 7th day, at 38,500 and 39,800 mg/L, respectively, while the other three groups reached their peak on the 12th day, at 33,300 (DO = 0.0 mg/L), 37,300 (DO = 0.4 mg/L), and 37,400 mg/L (DO = 0.8 mg/L), respectively (Figure 4(b)). This indicated that AP could convert most of the organic matter in the SM into soluble organic matter that could be utilised by microorganisms. Higher DO levels in the digesters result in faster conversion rates. This also confirmed the changing trend in the TCOD. According to Table 2, it could be observed that after AP, the SCOD concentrations in each group increased from 3.09% (DO = 0.4 mg/L) to 20.07% (DO = 2.0 mg/L) compared to that of the DO = 0.0 mg/L group. The results indicated that AP with different DO levels could enhance the hydrolysis of organic matter in the SM, and the hydrolysis enhancement of organic matter improved as the DO levels increased by promoting the biological and chemical processes of SM hydrolysis [26]. This was due to the brief aerobic environment that accelerated the hydrolysis of organic matter, thereby allowing rapidly biodegradable organic matter to be used in advance during subsequent AD. These results indicated that a microaeration environment could achieve a high methane yield and substrate degradation rate in AD systems by accelerating the degradation of organic compounds and shortening the reaction time required for hydrolysis and acidification [33].

3.3.3. Effect on VFAs

VFAs, the main intermediate products directly used by methanogens during digestion, are important indicators of the digestate during AD and directly affect pH fluctuations. In general, the inhibitory effect of VFAs on AD begins when acetic acid exceeds 2,000 mg/L or the TVFAs exceed 8,000 mg/L [27]. The concentration changes in TVFAs under different AP conditions are presented in Figure 5(a). According to changes in TVFAs concentration, the 37-day AD process could be divided into two stages. In Phase I (0–7 d), the TVFAs concentration gradually increased, whereas in Phase II (after 7

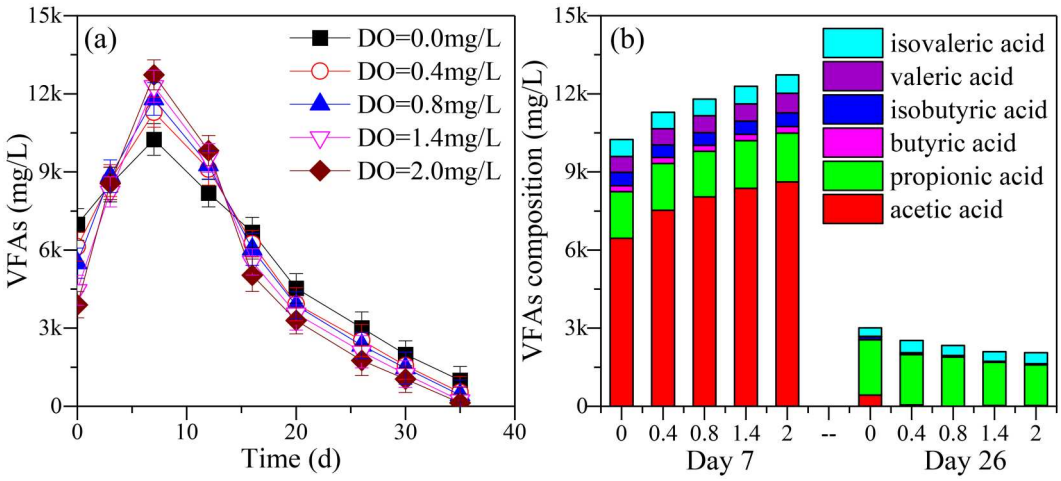


Figure 5. Changes in VFAs over time under different AP conditions.

d), the AD system entered the methane production stage, and the TVFA concentration rapidly decreased. According to Table 2, it could be observed that after AP, the VFAs concentrations in each group decreased from 12.31% (DO = 0.4 mg/L) to 44.24% (DO = 2.0 mg/L) compared to that of the DO = 0.0 mg/L group. Therefore, AP can alleviate acid inhibition during the early stages of the reaction. In Phase I, the TVFA concentration exhibited an upward trend under different DO levels and peaked on the 7th day. The TVFA concentration obtained in the DO = 2.0 mg/L group was the highest, with a maximum value of 12,727.13 mg/L, compared to that of the DO = 0.0 mg/L group, whereas TVFA production in the other groups increased by 10.18% (DO = 0.4 mg/L), 15.16% (DO = 0.8 mg/L), 19.95% (DO = 1.4 mg/L), and 24.22% (DO = 2.0 mg/L), respectively. These results revealed that AP promoted VFAs production during AD. This was similar to the results of Cao et al. [34], who observed that AP (1 L_{air}/L_{feed}/d) resulted in a 20.3% increase in the VFAs yield of agricultural wastes. This was primarily reflected in the rapid decrease in pH and rapid accumulation of VFAs. In Phase II, VFAs were rapidly consumed and gas production continued to increase. Methanogens began to rapidly use VFAs to produce methane, and the corresponding methane concentration increased. The pH of each group increased to different degrees during this stage.

Acetic acid was the main VFA component utilised in the AD of SM (Figure 5(b)). Acetic acid is considered a direct substrate utilised by methanogens. On the 7th day of the experiment, the VFAs reached their maximum values at different DO levels. The proportions of acetic acid at the maximum VFA concentrations were 62.95% (DO = 0.0 mg/L), 66.66% (DO = 0.4 mg/L), 68.18% (DO = 0.8 mg/L), 68.10% (DO = 1.4 mg/L), and 67.73% (DO = 2.0 mg/L). The results demonstrated that AP exerted no significant effect on the distribution of TVFA components in the early stage of the reaction ($p>0.05$), nor did it change the type of acid production metabolism of microorganisms in Phase I. In Phase II, acetic acid was rapidly consumed and converted into biogas. Compared to acetic acid, propionic acid was less easily converted to methane and remained relatively stable during Phase I and II. On the 7th day, the concentrations of propionic acid in each group ranged from 1,745.14 mg/L (DO = 0.8 mg/L) to 1,868.91 mg/L (DO = 2.0 mg/L). At the end of the reaction, the propionic acid concentrations in each group decreased from 2,120.45 (DO = 0.0 mg/L) to 1,564.63 mg/L (DO = 2.0 mg/L). Propionic acid-degrading microorganisms are the slowest growing and most sensitive acetic acid-producing organisms in most AD ecosystems. Oxygen can accelerate the acidification rate, improve the system stability, and prevent acid inhibition. Additionally, trace amounts of oxygen enhance electron transfer efficiency and microbial metabolism [35].

3.3.4. Effect on substance removal

TS and VS removal rates are important indicators of the degradation of digested material and the effectiveness of gas yield, and they are widely used to evaluate AD processes. The TS and VS removal rates of different digestion systems were calculated by comparing the TS and VS contents before and after digestion. The higher the TS and VS removal rates, the more substrate was used in the AD process. This helped reduce the number of fermentation residues.

The TS and VS removal rates in the batch digestion experiments are presented in Figure 6. The decreases

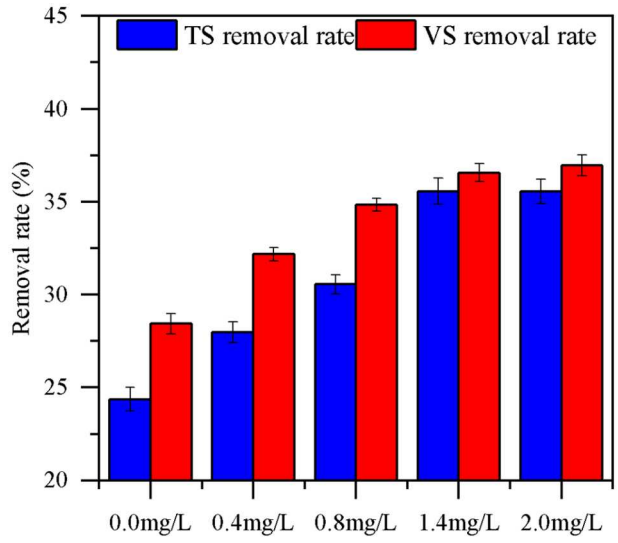


Figure 6. TS and VS removal efficiency under different AP conditions.

in the TS and VS removal rates in the AP group were greater than those in the untreated group. Compared to the DO = 0.0 mg/L group, AP increased the TS removal rate from 14.77% (DO = 0.4 mg/L) to 45.92% (DO = 1.4 mg/L), with the DO = 1.4 mg/L group being the highest. Compared to the DO = 0.0 mg/L group, AP increased the VS removal rate from 13.16% (DO = 0.4 mg/L) to 30.00% (DO = 2.0 mg/L), with the DO = 2.0 mg/L group being the highest. This was consistent with the TCOD removal rate. Research has demonstrated that AP can improve substrate utilisation efficiency and reduce digestion residues.

3.4. Kinetic study for methane yield

To evaluate the effect of AP on the AD of SM, improved Gompertz and logistic models [36] were used to analyze the dynamic simulation of the AD methane yield (Figure 7), and the dynamic parameters were calculated (Table 4). The results indicated that the dynamic parameters obtained from the improved Gompertz and logistic models exhibited similar relationships with different DO levels. The highest P_0 value was obtained in the DO = 2.0 mg/L group, indicating that SM could increase the methane yield by performing AD at an

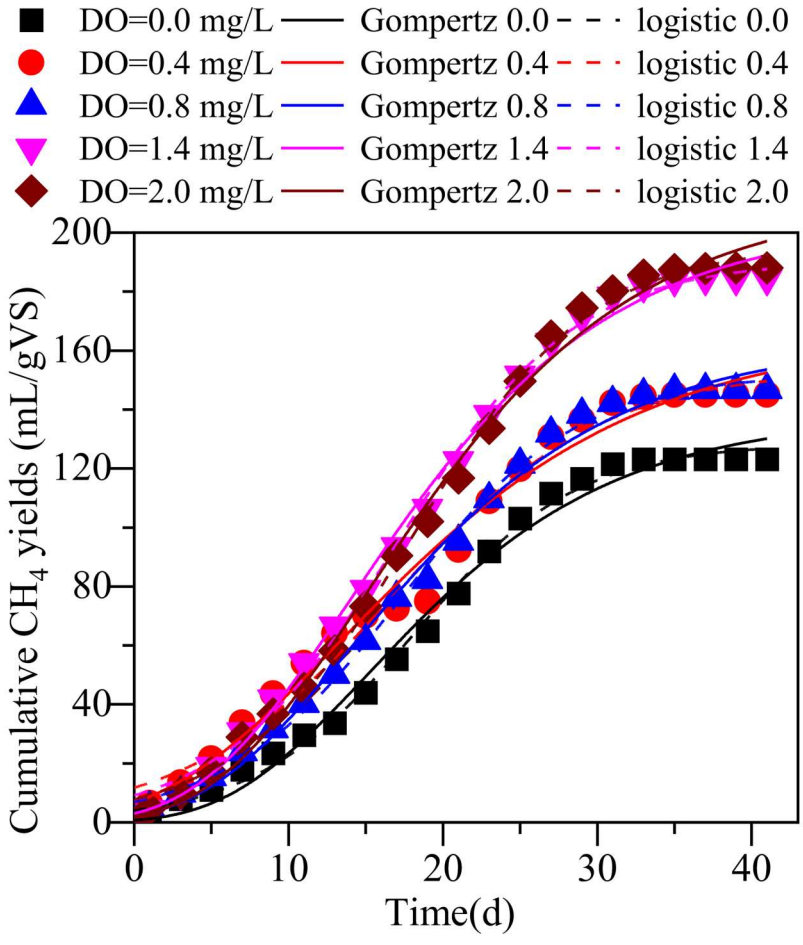


Figure 7. Measured and predicted methane yields using modified Gompertz model and logistic model.

Table 4. H_2S and sulfur concentrations in the biogas and digestates.

Item	Start sample	End sample				
		0.0 mg/L	0.4 mg/L	0.8 mg/L	1.4 mg/L	2.0 mg/L
H_2S in biogas (%)	–	0.412	0.349	0.306	0.234	0.209
Total sulfur in digestate (mg/g)	7.051	2.012	2.367	3.441	4.087	4.445

appropriate oxygen content. Among the groups undergoing AP, the R_{max} was significantly improved compared to that of the control group, with the DO = 2.0 mg/L group receiving the highest R_{max}. This was consistent with the data from P_0. During AD, the AP of raw materials can effectively accelerate the rate of methane yield. The lowest R_{max} was obtained in the DO = 0.0 mg/L group, which may be due to the accumulation of VFAs, causing a lower pH (Figure 3(a)) that may limit the conversion rate of SM.

In the DO = 2.0 mg/L group, λ was much smaller than that in the other groups, indicating that DO = 2.0 mg/L could improve the adaptability of microorganisms and shorten domestication time. T_{ef} can be used to predict organic matter conversion efficiency and methane yield. Compared to other groups, a maximum T_{ef} of 27.41 days and the highest methane yield were achieved in the DO = 2.0 mg/L group, indicating that the efficiency of organic matter conversion to methane was the highest in the DO = 2.0 mg/L group. The longer T_{ef} and shorter λ indicated that for the DO = 2.0 mg/L group, the AD efficiency of SM was higher and the process inhibition was smaller.

Additionally, the correlation coefficients (R^2) of the modified Gompertz and logistic models were both greater than 0.98, indicating that both models could be used for the dynamic simulation of AD in the AP. In the five experiments, the differences between the measured and predicted methane yields of the logistic model were lower than those of the modified Gompertz model (Table 4). Therefore, the logistic model was more accurate for fitting and predicting methane yield under different DO levels, and this is consistent with previous studies [37].

3.5. Kinetic study for reaction processes

Improved Gompertz and logistic models were used to calculate and simulate the effects of DO level on the AD reaction process.

The calculated and simulated reaction results are presented in Figure 8. Table 5 summarises the conversion potential (P_0') and maximum conversion rate (R_{max}') simulated using the two models. According to the R^2 values (Table 5), the modified Gompertz model exhibited better fitting performance than that of the logical model under different DO levels. Combining the best-fit results of the two models, the AP of AD in the SM not only accelerated the reaction rate but also increased the potential for hydrolysis, acidogenesis, acetogenesis, and methanogenesis. This significantly improved the AD of SM, accelerated the hydrolysis of macromolecular organic matter, reduced lag time, increased organic matter dissolution, and improved the maximum methane yield. In the four AD steps, except for the hydrogen and acetic acid production stages, P_0' gradually increased with increasing DO levels, and the highest P_0' was obtained in the DO = 2.0 mg/L group (Table 5). In terms of conversion potential at different stages, the P_0' values of acidogenesis, acetogenesis, and methanogenesis were similar for the DO = 1.4 mg/L group and the DO = 2.0 mg/L group. This indicates that most VFAs could be converted into acetate and then to methane by methanogens, leaving very few VFAs at the end of AD (Figure 5(b)). Meanwhile, it could be predicted from the P_0' order of methanogenesis, acetogenesis, and acidogenesis that due to the different reaction rates at each stage, VFAs may accumulate in other proportions. In the DO = 0.0 mg/L group, P_0' of acidogenesis was higher than that of the other stages, and this could easily lead to the accumulation of VFAs (Table 6).

The R_{max} values reflect the reaction rates of the four stages of the AD process. By comparing the R_{max}' values of four AD steps simulated by two models at DO = 0.0 mg/L with the order being R_{max}' (Meth.)> R_{max}' (Ace.)> R_{max} (Acid.)> R_{max} (Hydr.), and it could be observed that hydrolysis was the rate-limiting step in the use of SM in AD. With a continuous increase in the DO levels, R_{max}' increased, indicating that AP accelerated the hydrolysis rate in the AD process.

3.6. Microbial communities analysis

Microbial sequencing was performed at the final stage of the experiment using the biogas residue and feed SM. The sequencing results are presented in Figures 9 and 10.

3.6.1. Bacterial community

There were significant differences in the bacterial diversity between the AP and feed SM groups. The microbial diversity in the feed SM was relatively high; however, there were no dominant bacterial genera. There was

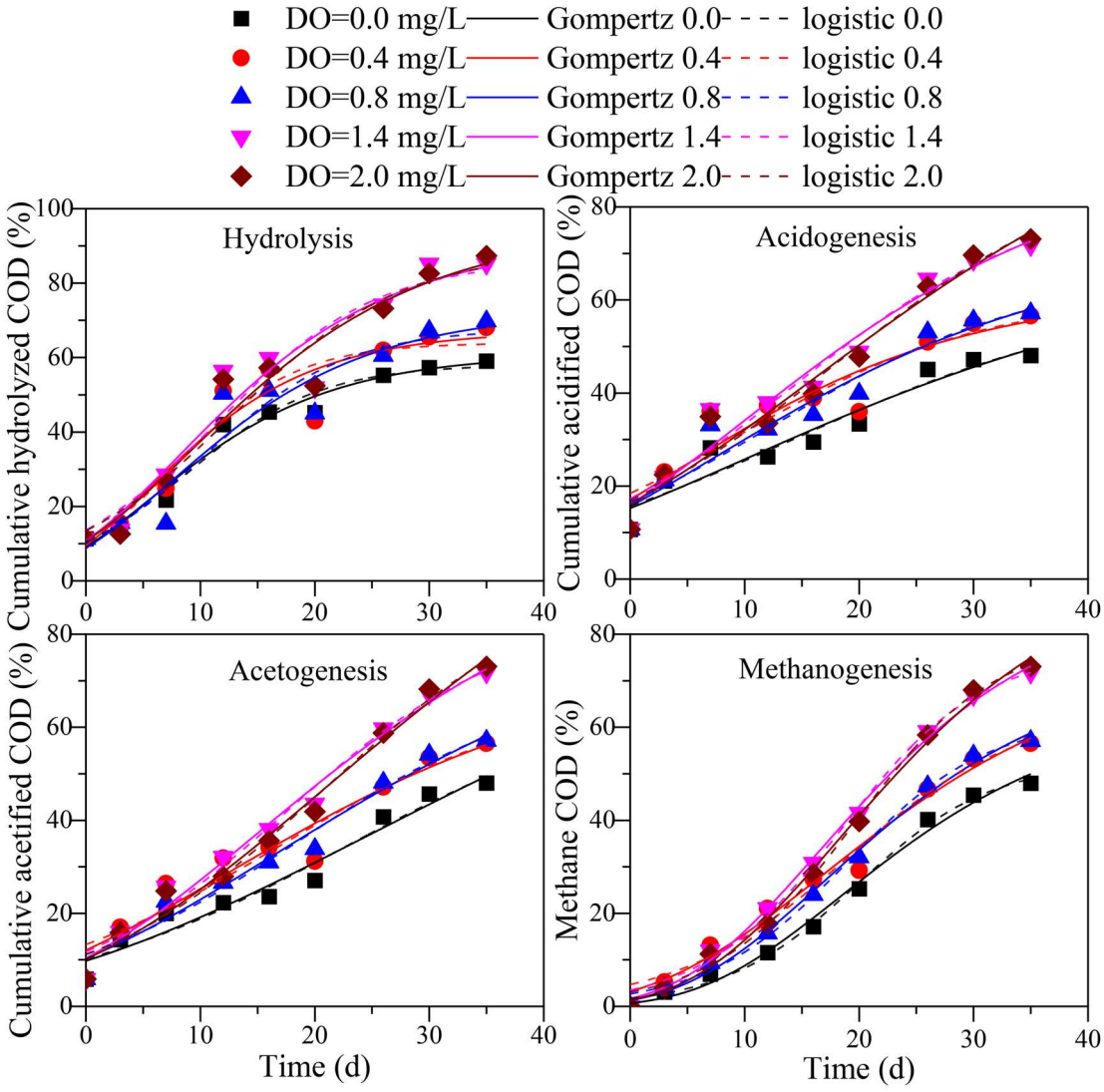


Figure 8. Description of the four steps in AD using modified Gompertz model and the logistic model.

no significant change in the bacterial community structure at the class level under different AP levels. Some important bacteria used for AP such as *Proteobacteria*, *Bacillus*, and *Staphylococcus*, were almost non-existent, whereas *Caldicoprobacter* was the predominant bacteria in all digestion tanks (Figure 9(a)), and the sum of their relative abundances accounted for 28.04% throughout the experiment. According to a previous study, *Caldicoprobacter* can metabolise difficult-to-degrade substances such as cellulose and hemicellulose into

Table 5. Kinetic parameters of methane production.

Model	DO level (mg/L)	Measured methane Yield (mL/gVS)	Predicted methane yield (P_0, mL/gVS)	R_{max} (mL/gVS/d)	Λ (d)	Tef (d)	R^2	Difference between measured and predicted methane yield (%)
Modified Gompertz model	0.0	123.10 ± 6.32	157.93 ± 9.07	5.28 ± 0.23	5.83 ± 0.52	23.87	0.992	28.29
	0.4	145.07 ± 5.41	188.76 ± 15.50	5.22 ± 0.27	4.87 ± 0.74	26.63	0.985	30.12
	0.8	146.48 ± 7.39	180.05 ± 6.23	6.00 ± 0.18	4.36 ± 0.36	27.14	0.996	22.92
	1.4	184.05 ± 2.36	221.90 ± 5.71	7.52 ± 0.17	4.12 ± 0.28	27.38	0.998	20.56
	2.0	187.47 ± 4.98	237.14 ± 8.37	7.63 ± 0.20	4.09 ± 0.32	27.41	0.997	26.49
Logistic model	0.0	123.10 ± 6.32	133.71 ± 2.63	5.96 ± 0.18	7.55 ± 0.34	22.15	0.997	8.62
	0.4	145.07 ± 5.41	161.59 ± 8.75	5.63 ± 0.34	5.38 ± 0.88	25.12	0.981	11.39
	0.8	146.48 ± 7.39	156.11 ± 3.03	6.63 ± 0.20	5.92 ± 0.36	25.58	0.997	6.57
	1.4	184.05 ± 2.36	194.14 ± 3.93	8.24 ± 0.26	5.62 ± 0.39	25.88	0.996	5.48
	2.0	187.47 ± 4.98	201.36 ± 3.86	8.49 ± 0.23	4.70 ± 0.33	26.80	0.997	7.41

Table 6. Kinetic parameters of each AD step in the modified Gompertz model and logistic model.

DO level (mg/L)	Parameter	Modified Gompertz model				Logistic model			
		Hydrolysis	Acidogenesis	Acetogenesis	Methanogenesis	Hydrolysis	Acidogenesis	Acetogenesis	Methanogenesis
0.0	P_0' (%)	60.97 ± 3.64	68.16 ± 28.05	99.86 ± 67.94	64.65 ± 8.24	58.12 ± 2.64	59.47 ± 15.84	68.21 ± 21.88	52.61 ± 2.26
	R_{max}' (%/d)	0.91 ± 0.34	1.08 ± 0.24	1.26 ± 0.18	1.95 ± 0.15	0.90 ± 0.34	1.08 ± 0.19	1.31 ± 0.15	2.25 ± 0.13
	R^2	0.967	0.903	0.945	0.988	0.966	0.900	0.944	0.994
0.4	P_0' (%)	67.22 ± 6.92	60.96 ± 12.39	76.52 ± 28.47	76.89 ± 13.48	63.92 ± 5.10	59.75 ± 11.18	68.01 ± 18.50	63.79 ± 6.94
	R_{max}' (%/d)	1.50 ± 0.82	1.59 ± 0.54	1.44 ± 0.28	1.95 ± 0.17	1.41 ± 0.87	1.39 ± 0.43	1.45 ± 0.25	2.13 ± 0.23
	R^2	0.886	0.850	0.907	0.978	0.879	0.834	0.897	0.973
0.8	P_0' (%)	72.66 ± 10.48	70.80 ± 16.85	88.89 ± 26.98	74.19 ± 6.12	67.75 ± 6.90	65.05 ± 11.32	71.12 ± 12.50	61.87 ± 2.41
	R_{max}' (%/d)	1.68 ± 0.70	1.47 ± 0.32	1.52 ± 0.13	2.23 ± 0.12	1.75 ± 0.73	1.43 ± 0.27	1.60 ± 0.16	2.52 ± 0.14
	R^2	0.887	0.921	0.965	0.994	0.880	0.914	0.962	0.995
1.4	P_0' (%)	91.59 ± 12.21	91.56 ± 19.72	95.43 ± 18.28	90.28 ± 5.45	86.22 ± 9.10	81.85 ± 12.25	84.04 ± 9.59	76.55 ± 2.95
	R_{max}' (%/d)	2.15 ± 0.67	1.91 ± 0.30	2.05 ± 0.16	2.80 ± 0.13	2.10 ± 0.69	1.92 ± 0.28	2.17 ± 0.19	3.14 ± 0.18
	R^2	0.924	0.948	0.980	0.996	0.910	0.942	0.976	0.995
2.0	P_0' (%)	95.26 ± 13.45	113.35 ± 40.09	117.53 ± 37.92	97.40 ± 8.22	88.48 ± 9.72	91.51 ± 18.56	93.10 ± 13.74	79.53 ± 2.97
	R_{max}' (%/d)	2.99 ± 0.55	1.82 ± 0.19	2.09 ± 0.13	2.86 ± 0.14	2.98 ± 0.59	1.90 ± 0.21	2.23 ± 0.16	3.25 ± 0.16
	R^2	0.939	0.953	0.979	0.995	0.925	0.951	0.978	0.996

oligosaccharides or monosaccharides that can be utilised by microorganisms [38]. Compared to the control group, the number of bacteria in the other groups exhibited varying degrees of increase. The abundance of *Caldicoprobacter* in the DO = 2.0 mg/L group was 34.96%, which was 40.12% higher than that in the control group. In addition to the main bacteria mentioned above, other bacteria detected in these digests included *HN-HF0106*, *Clostridium sensu stricto 1*, *Ruminiclostridium*, and *Syntrophomonas*. Finally, many bacteria such as *Clostridium Sensu Stricto 1* have been bred to degrade lignocellulose under extremely low-oxygen conditions [39]. *Syntrophomonas* is an efficient propionic acid-oxidising bacterium that can metabolise VFAs [40] and construct a direct interspecies electron transfer (DIET) mechanism with specific methanogens [41]. Additionally, the abundance of *Bacteroidales* decreased to varying degrees after AP. *Bacteroidetes* use the *buk* pathway, whereas *Clostridium* can produce butyric acid through the *buk* pathways [42]. Additionally, *Bacteroides* utilises anaerobic respiration for glycolysis to produce acetic and isovaleric acids [43]. It is one of the anaerobic bacteria with the highest oxygen resistance and maintained a relatively low abundance throughout the AD process in this study. There were no significant differences in the dominant bacteria at different DO levels ($p > 0.05$).

3.6.2. Archaea community

The diversity of the archaea in each group improved slightly after AP (Figure 9(b)). *Methanobacterium*, *Methanosarcina* and *Methanosaeta* were the primary methanogens in all the groups, regardless of the amount of oxygen used for pretreatment. The abundances of archaea in *Metanobacterium, Metanosarcina*, and *Metanosaeta* in the DO = 0.0 mg/L group were 56.73%, 14.73%, and 12.54%, respectively. Compared to the DO = 0.0 mg/L group, the abundance of archaea in *Metanobacterium* in each group treated with micro oxygen exhibited varying degrees of decrease, with the DO = 0.8 mg/L group exhibiting the largest decrease at 46.05%. The content change of *Methanobacterium* was the highest. *Methanobacterium* is an archaea that uses methane as a specific metabolite and is key archaea in smooth AD and methane production processes. By comparing these two genera in the same microbial agent group, it was observed that the abundance of archaea in *Metanosarcina* in the control group was the lowest, whereas the AP group exhibited an increase compared to the other groups, and the relative abundance of *Metanosaeta* exhibited a relatively gentle trend. For example, in the DO = 0.0 mg/L group, the relative abundances of *Metanosarcina* and

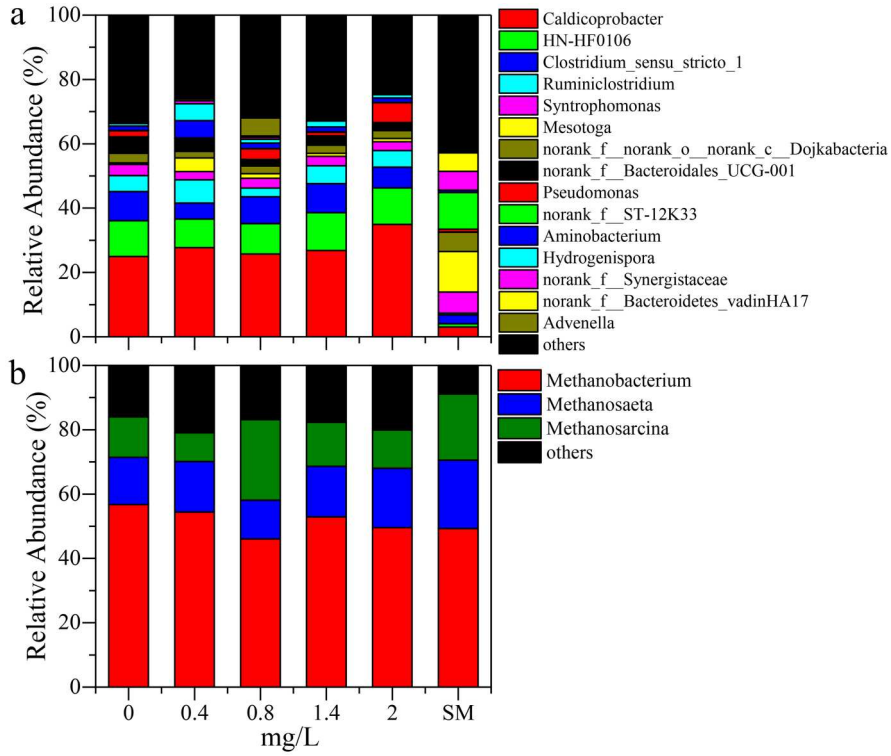


Figure 9. Bacterial community structures (a) and Archaeal community structure (b) at the genus level under different AP conditions.

Metanosaeta were 12.54% and 14.73%, respectively. At the DO = 0.8 mg/L group, *Metanosarcina* increased significantly, reaching 25.06%, whereas *Metanosaeta* remained at 12.05%. *Metanosarcina* is considered to be the only methanogenic bacterium compatible with the acetylation and hydrogen nutrition pathways, whereas *Metanosaeta* can only follow the acetylation pathway [44]. The transition between *Metanosarcina* and *Metanosaeta* may be due to changes in the precursors produced by *Metanosarcina*. Under different DO levels, there were no significant differences in the dominant archaea ($p > 0.05$).

3.6.3. Correlation between microbial community and environment factors

RDA was conducted on the microbial community structure and AD environmental factors to reveal the influence of environmental factors on the microbial community structure under different AP intensities. The results are presented in Figure 10.

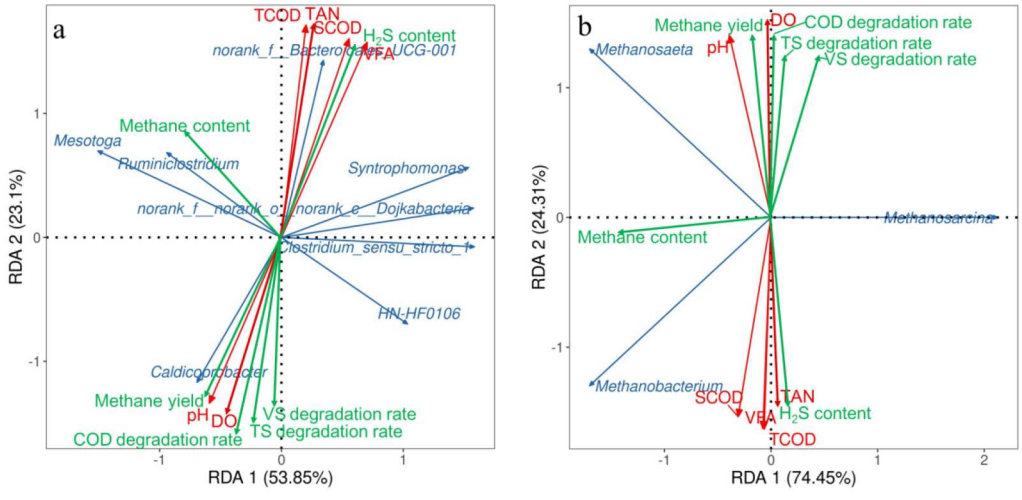


Figure 10. Redundancy analysis (RDA) between microorganisms and environmental factors.

Figure 10(a) presents the correlation between bacterial genera and environmental factors. *Clostridium Sensu Stricto 1* was positively correlated with the concentrations of various organic compounds (including TCOD, SCOD, and VFAs) and TAN. This indicated that *Clostridium Sensu Stricto 1* possessed strong hydrolytic activity and good tolerance towards substrates and that organic acids and TAN may promote an increase in its relative abundance. *Caldicobacter* positively correlated with methane yield and organic matter removal efficiency. This indicated that *Caldicobacter* could fully utilise the hydrolysed products and achieve substance conversion. Additionally, *Caldicobacter* is more suitable for living in environments with high DO levels and is more sensitive to inhibitors (VFAs and TAN). Therefore, the abundance of *Caldicobacter* increased significantly after AP. According to previous reports, *Caldicobacter* was the dominant microorganism in AD systems and plays an important role in biogas yield [45].

Figure 10(b) presents the correlation between the archaeal genera and environmental factors. *Methanosaeta* exhibits a strong positive correlation with methane yield, DO levels, and pH. This result indicates that *Methanosaeta* was the main methane-producing archaea and had a high tolerance to oxygen. However, *Methanosaeta* was significantly and negatively correlated with VFAs and TAN, indicating poor stability and susceptibility to inhibition. This was similar to the results of previous research [46]. *Methanobacterium* was significantly correlated with TAN and VFAs but negatively correlated with DO levels. This indicated that *Methanobacterium* possessed strong adaptability to ammonia inhibition and acid inhibition; however its activity decreased at high oxygen concentrations. Therefore, *Methanobacterium* was highly abundant in all experimental groups but exhibited a decreasing trend with increasing DO levels. A similar finding was reported by Cheng et al. [47], who demonstrated that the methanogens in the AP group were primarily *Methanosarcina*, whereas those in the non-pretreatment group were primarily *Methanobacterium*.

The RDA results were consistent with the composition and distribution of microbial genera, thus indicating that environmental factors had a significant impact on the distribution of microbial communities. In practical applications, it is necessary to control key environmental factors to obtain an efficient methane-producing microbial community structure and enhance the methane yield potential of the substrates.

4. Conclusions

Microaerobic bacteria improved methane yield during SM batch experiments by regulating the microbial community structure. Facultative bacteria were capable of secreting hydrolytic enzymes and were stimulated by oxygen, leading to more rapid hydrolysis and higher VFA yield. Methanogens with oxygen tolerance were also enriched. This study demonstrated the beneficial effects of AP on batch AD of SM, which could also provide a reference for the industrial application of aeration technology.

Acknowledgments

We would also like to thank Editage (www.editage.cn) for English language editing.

Disclosure statement

No potential conflict of interest was reported by the author(s) .

Funding

This work was supported by Young Scientific and Technological Talents at Universities of Inner Mongolia Autonomous Region: [grant number NJYT22083]; The Project of Inner Mongolia 'Prairie Talents' Engineering Innovation Entrepreneurship Talent Team.

Data availability statement

The data that support the findings of this study are available from the corresponding author, [Benyi Xiao], upon reasonable request.

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